

Engineering Ethics

“Code of Ethics”

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Agenda for today

1. Introduction
2. The “Shell Decision”
3. How to decide?

The Unknown



The Unknown

- SARS
- Bird flu
- Ozone depletion
- Harmful substances
- Global warming
- Climate change
- Covid-19
- Migration
- Chernobyl
- Ebola
- BSE
- Nuclear radiation
- Sweetwater scarcity
- Resource scarcity
- Rising sea levels
- Methane
- Cigarette smoke
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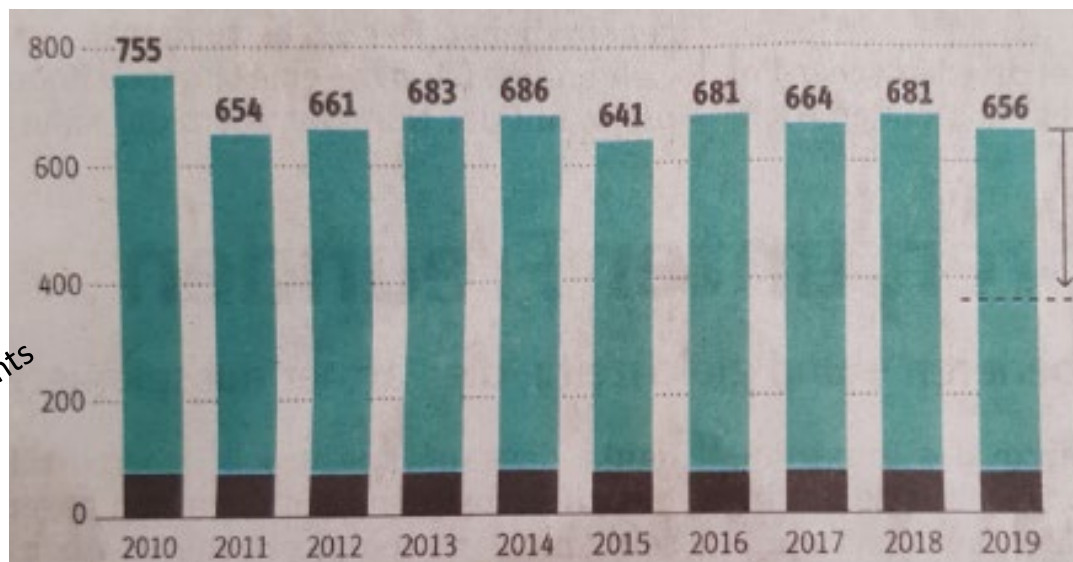
Read the IEEE-Ethics
Codex:

Which of these
unknown are covered
by the Codex - give
reasons

The „Shell Decision“

- What is the consequence of the decision?

- Direct emissions
- indirect emissions
- Emissions coming from the use of Shell's products



Shell is to lower its emissions by 45% by 2030 compared to 2019.

Shell wants to be climate neutral by 2050

The „Shell Decision“

- What happens in the „legal field“
 - The courts pass the balls to each other
 - The German Constitutional Court cites a Dutch decision
 - The Hague “Urgenda judgement” of 2019, which ordered the Netherlands to reduce CO₂
 - The German Constitutional Court increases media range by publishing its decision in
 - German
 - English
 - French and
 - Spanish

The „Shell Decision“

- Who is harmed by climate change and how?
 - The Dutch are living at the waterfront
 - 10,000 + x people dying because of emissions from traffic

The „Shell Decision“

- In essence, the court justifies the reduction obligation for Shell with rules on liability for damages that are familiar from everyday life.
 - What is innovative about this is that the court derives a special duty of care from it: The company's duty to reduce damage caused by climate change.
 - The claim of compensation is directed to future damages and their prevention.
 - Based on the legally guaranteed protection of life and the right to respect for private life from the European Convention on Human Rights

The „Shell Decision“

- Would a similar decision be possible in Germany?
- Yes !
- Legal Duty to Maintain Safety (Verkehrssicherungspflicht)
 - If you operate a bouncy castle, you have to make sure that nothing happens during use

Coming to the Ethical Standards and to Ethics Codes

How to decide ?

First Standard: The Golden Rule

„«Would I be willing to accept the effects of this practice?» This question is especially difficult to answer when it requires putting myself in the position of a host country citizen, whose culture, economic status, living conditions, and values may be very different from my own.“

- An example: „Supply-chain law“ (Germany, May 2021) - A supply chain law should oblige or hold companies liable to trace intermediate goods or finished products procured abroad at all stages of their supply chain for any production processes that are harmful to the environment or violate labour conditions.

How to decide ? A drastic example



The sulfure miner at Ijen (Indonesia)

<https://www.youtube.com/watch?v=Gv6g3w8G18>

How to decide ?

Second Standard: Universal Human Rights

The rights to:

- Life
- Liberty
- Security of Person
- Recognition before the law
- An impartial trial
- Marriage
- Property ownership
- Freedom of thought
- Peaceful assembly and participation in government
- Social security and work
- education
- Participate in and form trade unions
- Non-discrimination
- A minimal standard of living

Now to decide ?

Third Standard: Promoting Basic Human Well-Being I

- Being able to live a human life of normal length.
- Being able to enjoy good health, nourishment, shelter, sexual satisfaction and physical movement.
- Being able to avoid unnecessary and non-beneficial pain and to have pleasurable experiences.
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Now to decide ?

Third Standard: Promoting Basic Human Well-Being II

- Being able to form a conception of the good and to engage in critical reflection about the planning of one's life.
- Being able to show concern for others and to engage in social interaction.
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- Being able to life one's own life and nobody else's.

Now to decide ?

Fourth Standard: Ethics Codes

- Almost every Association of Professional Engineers will have a Code of Ethics
- The Codes of Ethics are similar for
 - Civil Engineers
 - Electrical Engineers
 - Mechanical Engineers

Now to decide ?

"Engineers uphold and advance the integrity, honor and dignity of the engineering profession by:

1. using their knowledge and skill for the enhancement of human welfare;
2. being honest and impartial, and serving with fidelity their clients (including their employers) and the public; and
3. striving to increase the competence and prestige of the engineering profession."

How to decide

- You heard about Whistleblowers
 - Download and read the ASME Code of Ethics (in moodle)
- Compare the general information with the Code of Ethics
 - What rules did a whistleblower (possibly) break?
 - The consideration of which rules supported whistleblower actions?